1 **(a)** $2(2x+3)+2(x-1)=34$ OR $2x+3+x-1=17$ **[M1]**

$6x+4=34$ OR $3x+2=17$ **[M1]**

$x=5$ **[A1]**

(b) Substitution of $x=5$: $(2(5)+3)\times(5-1)$ OR 13×4 **[M1]**

52 **[A1]**

[Total Marks: 5]

2 **(a)** $\frac{1}{2}(10+16)\times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle $=16-10=6$ **[M1]**

Slanted edge $=\sqrt{6^2+8^2}=10$ **[M1]**

$8+10+16+10=44$ **[A1]**

[Total Marks: 5]

3 **(a)** $\frac{1}{2}\times(15+24)\times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A=\sqrt{10^2-8^2}=6$ **[M1]**

Horizontal distance for the right-angled triangle under $B=24-15-6=3$ **[M1]**

$BC=\sqrt{8^2+3^2}$ $(=\sqrt{73}\approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Total Marks: 6]

4 **(a)** Area of rectangle $=4(2x+3)$ OR $8x+12$ **[M1]**

Area of triangle $=\frac{1}{2}\times 6\times(3x-1)$ OR $9x-3$ **[M1]**

$8x+12=9x-3$ **[M1]**

$x=15$ **[A1]**

(b) Length $=2(15)+3=33$ **[M1]**

Perimeter $=2(4+33)=74$ **[A1]**

[Total Marks: 6]

5 **(a)** $\frac{1}{2}(x+x+4)(x-1)=40$ **[M1]**

$\frac{1}{2}(2x+4)(x-1)=40$ OR $(x+2)(x-1)=40$ **[M1]**

$x^2+2x-x-2=40 \implies x^2+x-42=0$ **[A1]**

(b) $(x+7)(x-6)=0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top = 6, Bottom = 10, Height = 5 **[M1]**

Horizontal difference = 4, so slanted side = $\sqrt{5^2+4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter = $5+6+10+\sqrt{41}$ **[M1]**

27.4 (or 27.40...) **[A1]**

[Total Marks: 9]

6 (a) Perimeter of rectangle = $6y$ OR Perimeter of square = $60-6y$ **[M1]**

$\frac{60-6y}{4}$ OR $15-1.5y$ **[A1]**

(b) $(15-1.5y)^2+y(2y)=113$ **[M1]**

$225-45y+2.25y^2+2y^2=113$ **[M1]**

$4.25y^2-45y+112=0$ leading to $17y^2-180y+448=0$ with no errors seen **[A1]**

(c) $\frac{-(-180)\pm\sqrt{(-180)^2-4(17)(448)}}{2(17)}$ **[M1]**

$\frac{180\pm\sqrt{1936}}{34}$ OR $\frac{180\pm44}{34}$ **[M1]**

$y=4$ and $y=6.59$ (or 6.588... or $\frac{112}{17}$) **[A1]**

(d) Identifies $y=4$ as the valid solution (since $15-1.5(6.59)<6.59$) leading to side length = 9 **[M1]**

81 **[A1]**

[Total Marks: 10]

7 (a) Calculates area of triangle: $\frac{1}{2}\times12\times8=48$ **[M1]**

$4x=48$

x = 12[A1]

(b) Uses Pythagoras to find slanted side: $\sqrt{6^2+8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter = $12+10+10=32$ **[A1]**

[Total Marks: 5]

8 (a) Uses Area = $\frac{1}{2}\times$ base $\times$ height = $\frac{1}{2}\times14\times9$ **[M1]**

63 **[A1]**

(b) Finds $AE=\sqrt{9^2+9^2}=\sqrt{162}$ or 12.7... **[M1]**

Finds $BE=\sqrt{5^2+9^2}=\sqrt{106}$ or 10.2... **[M1]**

Perimeter $= 14 + \sqrt{162} + \sqrt{106}$ **[M1]**

37.0 (or 37.02 . . .) **[A1]**

[Total Marks: 6]

9 (a) (i) Sets up equation for area: $\frac{1}{2}((x+4)+(3x-2))(x-1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x+2)(x-1) = 20$ or $(2x+1)(x-1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

(ii) Factorises $(2x-7)(x+3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side CD: $\sqrt{(8.5-7.5)^2 + 2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2 + 2.5^2} = \sqrt{7.25}$ (or 2.69 . . .) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69\cdots = 21.2$ (or 21.19 . . .) **[A1]**

[Total Marks: 8]